📘 CUET Computer Networks (CSE-311/411) — Complete 5-Year Solutions (2019–2023)

> *Every unique question from the last 5 years, solved once. Repeated questions are merged.* ***Sources:*** *Rifat PDF (page refs), materials/slides/, materials/books/*

PART A: NETWORK BASICS & PHYSICAL LAYER

Q: OSI vs TCP/IP Model — Compare and Contrast

Asked: 2023 Q.1a, 2022 Q.2a (layers), 2021 Q.2a (TCP/IP layers), 2020 Q.2a, 2019 Q.1a **Rifat PDF: Pages 1, 4 | Slides: OSI_and_TCP_IP_Layering_Models.pdf**

Feature	OSI Model (7 Layers)	TCP/IP Model (4 Layers)
Development	Theoretical model by ISO, defined before protocols	Practical model by DoD, built around existing protocols
Layers	Application, Presentation, Session, Transport, Network, Data Link, Physical	Application, Transport, Internet, Network Access
Network Layer	Supports both connectionless and connection-oriented	Supports only connectionless (IP)
Transport Layer	Supports only connection-oriented	Supports both TCP (connection) and UDP (connectionless)
Protocol Coupling	Protocol-independent	Tightly coupled with TCP/IP suite

Why TCP/IP is more popular: It was built organically alongside real networks (ARPANET/UNIX). OSI was academic and arrived too late. TCP/IP's 4-layer simplicity maps directly to how software is programmed.

Encapsulation forms layers: each layer receives data from above, treats it as payload, adds its own header. This creates strict boundaries — a Transport protocol need not understand Network layer internals.

Q: What is Computer Network? Components? Internet as "Network of Networks"

Asked: 2021 Q.1a, 2020 Q.1a **Rifat PDF: Pages 1–3**

Computer Network: A collection of autonomous computers interconnected by a single technology, sharing resources using standard protocols.

Components: (1) End Devices (PCs, phones), (2) Network Devices (switches, routers), (3) Transmission Media (twisted pair, fiber, wireless), (4) Protocols (TCP/IP, Ethernet).

Internet = "Network of Networks": The Internet is not one network — it interconnects thousands of independent LANs, WANs, and ISPs using TCP/IP as the common glue.

Q: Network Topologies — Types, Pros, Cons

Asked: 2023 Q.1c/Q.2a, 2022 Q.2a, 2021 Q.1b, 2020 Q.1b, 2019 Q.1b **Rifat PDF: Pages 6, 11, 14, 15 | Slides: Network_Topology_Overview.pdf, LAN_Technology_and_Network_Topologies.pdf**

Topology	Pros	Cons
Bus	Simple, least cabling	Backbone break kills all; heavy collisions
Star	Robust (one cable break = one PC down); easy troubleshooting	Central hub = single point of failure; intensive cabling
Ring	Token passing = zero collisions; fair access	Ring break halts entire network
Mesh	Highly fault-tolerant; no single point of failure	Expensive: n(n-1)/2 links needed
Tree/Hybrid	Scalable; hierarchical grouping	Complex; inherits star's central-point weakness

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Q: Protocols and Topologies Definitions; Logical Topologies

Asked: 2022 Q.2a, 2019 Q.1b

- **Protocol:** A set of rules governing data format, timing, sequencing, and error control.
- **Physical Topology:** Actual cable layout. **Logical Topology:** How data flows conceptually.
- **Logical Ring (Token Ring):** Deterministic, zero collisions, fair. Slow token overhead.
- **Logical Bus (Ethernet CSMA/CD):** Fast under light loads; collisions surge under heavy load.

Q: Unicast, Broadcast, Multicast

Asked: 2020 Q.1b

- **Unicast:** One-to-one (single sender to single receiver).
- **Broadcast:** One-to-all (every device on the subnet receives it). Example: ARP Request.
- **Multicast:** One-to-many (only subscribed group members receive). Example: IPTV.

Q: Wired vs Wireless Networking

Asked: 2021 Q.1c, 2020 Q.1c **Rifat PDF: Pages 58–59**

- **Wired:** Guided media (Cat6, Fiber). High bandwidth (100 Gbps on fiber), low latency, physically secure. No mobility.
- **Wireless:** Unguided (Wi-Fi, 5G). Full mobility. Suffers attenuation, interference, needs encryption (WPA3).

Q: Collision Domain vs Broadcast Domain

Asked: 2023 Q.1c, 2021 Q.1c, 2020 Q.1c, 2019 Q.5a **Rifat PDF: Pages 61–67, 72 | Slides: Hardware_Hub_Bridge_Router_Switch_QA.pdf**

- **Collision Domain:** Segment where simultaneous transmissions collide. Hubs = one big domain. Each switch port = separate domain.

- **Broadcast Domain:** Segment where a Layer-2 broadcast reaches all hosts. Switches forward broadcasts. Only routers block them.
- **Situation where broadcast is necessary:** ARP — a host broadcasts "Who has IP 192.168.1.5?" to discover the MAC address.

Q: Hub, Switch, Router — Construction and Operation

Asked: 2021 Q.1d, 2020 Q.1c/d, 2019 Q.8c **Rifat PDF: Pages 14, 62, 72 | Slides: Computer_Networking_Devices_Overview.pdf**

- **Hub (Layer 1):** Dumb multiport repeater. Broadcasts to ALL ports. Creates one collision domain.
- **Switch (Layer 2):** Learns MAC addresses, forwards frames only to the correct port. Eliminates collisions.
- **Router (Layer 3):** Routes between different IP networks using routing tables. Breaks broadcast domains.

Q: Token Passing and FDDI

Asked: 2021 Q.1d, 2019 Q.8b **Rifat PDF: Page 73 | Slides: LAN_Technologies_Must_Read.pdf**

- **Token Passing:** Deterministic access method. A "Token" frame circulates the ring. Only the token holder can transmit. Zero collisions by design.
- **FDDI (Fiber Distributed Data Interface):** 100 Mbps fiber-optic LAN standard using a dual counter-rotating ring with token passing. If a cable is cut, the two rings "wrap" into a single loop — providing fault tolerance.

Q: Shannon's Theorem & Nyquist Theorem

Asked: 2023 Q.2b/c, 2019 Q.2d, CSE-411 Q.3c **Rifat PDF: Page 23 | Slides: Nyquist_and_Shannon_Theorems.pdf, Shannon_Channel_Capacity.pdf**

Nyquist (Noiseless): $C = 2 \times B \times \log_2(L)$ where B=bandwidth, L=signal levels. **Shannon (Noisy):** $C = B \times \log_2(1 + SNR)$ — gives the theoretical maximum.

Solved: Bandwidth = 4300 Hz (2019 Q.2d: C=8600, L=2 → B = 8600/2 = 4300 Hz) **Solved: Bandwidth = 2000 Hz** (2019 Q.2d variant: C=4000, S/N=3 → B = 4000/$\log_2$(4) = 2000 Hz) **Solved: C ≈ 10,656 bps** (CSE-411: B=1600, SNR=20dB → SNR_linear=100, C=1600×$\log_2$(101)≈10,656)

PART B: DATA LINK LAYER

Q: Framing — Definition and Methods

Asked: 2023 Q.3a, 2021 Q.2a, 2020 Q.2b, 2019 Q.2b **Rifat PDF: Pages 6, 8, 32–35 | Slides: Data_Link_Framing_and_Error_Control.pdf**

Framing: The Layer-2 process of breaking a raw bit stream into discrete logical "frames" so the receiver knows where a message starts and ends.

Four Methods: (1) Character Count, (2) Character/Byte Stuffing, (3) Bit Stuffing, (4) Physical Layer Coding Violations.

- **Character Count:** Frame length embedded in header. Simple but brittle — a single bit error in the count field loses sync permanently.
- **Byte Stuffing:** Special FLAG byte marks boundaries. If FLAG appears in data, an ESC byte is stuffed before it.

- **Bit Stuffing:** Flag = `01111110`. If five consecutive `1`s appear in data, a `0` is stuffed after them. Receiver removes it automatically.

Destuffing Example (2023 Q.3b): Data `0110 0111 1100 1111 0111 1101` → Find five `1`s followed by `0`: remove the stuffed `0` after sequences of five `1`s.

Q: Flow Control — Sliding Window Protocol

Asked: 2023 Q.3b, 2021 Q.2c, 2020 Q.3a **Rifat PDF: Pages 54, 55, 119 | Slides: Data_Link_Flow_and_Error_Control.pdf**

Why Sliding Window? Stop-and-Wait is inefficient — the sender idles 90% waiting for ACKs. Sliding Window allows sending W frames before needing an ACK, keeping the pipe full.

Go-Back-N vs Selective Repeat:

- **Go-Back-N:** Receiver rejects out-of-order frames. If frame 3 is lost, frames 4,5,6 are discarded and sender re-sends 3,4,5,6. Simple but wasteful.
- **Selective Repeat:** Receiver buffers out-of-order frames. Only the lost frame is retransmitted. Bandwidth efficient but complex buffering.

Q: CSMA/CD and CSMA/CA

Asked: 2023 Q.1c, 2022 Q.3b, 2021 Q.2a, 2020 Q.3d, 2019 Q.1c **Rifat PDF: Pages 61 | Slides: Ethernet_CSMACD_Exponential_Backoff.pdf, Wireless_MAC_Protocols.pdf**

- **CSMA/CD (Collision Detection — Wired Ethernet):** Listen → Transmit → Monitor voltage. If spike detected (collision) → Abort → Send Jam Signal → Binary Exponential Backoff → Retry.
- **CSMA/CA (Collision Avoidance — Wireless 802.11):** Cannot detect collisions while transmitting. Uses RTS/CTS handshake to reserve the channel before sending.

1-persistent vs Non-persistent CSMA:

- **1-persistent:** If channel busy, keep listening; transmit immediately when idle. Aggressive — high collision risk.
- **Non-persistent:** If busy, wait a random time before listening again. Reduces collisions but adds latency.

Q: CRC Checksum Calculations

Asked: 2023 Q.3a, 2022 Q.2c, 2020 Q.2d, 2019 Q.2c, CSE-411 Q.3a **Rifat PDF: Pages 21, 135, 140 | Slides: Packets_Frames_and_Error_Detection.pdf**

Solved: Frame `101101110011`, $G(X) = X^4+X^2+1$ → Binary `10101`

1. Append 4 zeros: `1011011100110000`
2. Modulo-2 divide by `10101`
3. Remainder = CRC checksum → Append to original frame

Solved: Frame `101100101`, $G(x) = x^3+x^2+1$ → Binary `1101`

1. Append 3 zeros: `101100101000`
2. Modulo-2 division:

```
          111101100
     ______________
1101 | 101100101000
       1101
       ----
        1100
        1101
        ----
          0110
          0000
          ----
           1101
           1101
           ----
             0001
             0000
             ----
               0010
               0000
               ----
                 0100
                 0000
                 ----
                  1000
                  1101
                  ----
                   101 (Remainder)
```

Transmitted Frame: `101100101101`

Solved: Frame `10101011011`, Generator `10001` — Verify reception of `10101011010`

Divide received `10101011010` by `10001`. Remainder = `1010` ≠ 0. **NOT accurate reception — errors occurred.**

Solved: Data `10011011100`, G(x) = x^4+x^2+1 → `10101`

Append 4 zeros → `100110111000000`. After division, Remainder = `0100`. **Transmitted Codeword:** `100110111000100`

Q: Hamming Code — Error Correction

Asked: 2023 Q.2c ("CAFE"), 2021 Q.2d ("SEE"), 2020 Q.3c ("CSE"), 2019 Q.4c ("CUET"), CSE-411 (data 1010) **Rifat PDF: Pages 42, 46 | Slides: Data_Link_Framing_and_Error_Control.pdf**

Method: For 7 data bits → 4 parity bits (positions 1,2,4,8). Total = 11 bits. Format: P_1, P_2, D_3, P_4, D_5, D_6, D_7, P_8, D_9, D_{10}, D_{11}

Letter	ASCII (7-bit)	Hamming Code (11-bit)
C (1000011)	D=1,0,0,0,0,1,1	`01100000011`
A (1000001)	D=1,0,0,0,0,0,1	`11100000001`

F (1000110)	D=1,0,0,0,1,1,0	`01100000110`
E (1000101)	D=1,0,0,0,1,0,1	`10100000101`
S (1010011)	D=1,0,1,0,0,1,1	`00110100011`
U (1010101)	D=1,0,1,0,1,0,1	`11110100101`
T (1010100)	D=1,0,1,0,1,0,0	`00110101100`

Solved: 4-bit data `1010` (CSE-411)

3 parity bits needed. Format: $P_1,P_2,D_3,P_4,D_5,D_6,D_7$. Data: $D_3=1$, $D_5=0$, $D_6=1$, $D_7=0$. $P_1(1,3,5,7)=1$, $P_2(2,3,6,7)=0$, $P_4(4,5,6,7)=1$. **Codeword:** `1011010` Error correction: If bit 5 flips → Syndrome = $P_4P_2P_1 = 101_2 = 5$ → flip bit 5 back.

PART C: TRANSPORT LAYER

Q: TCP vs UDP

Asked: 2023 Q.3c, 2022 Q.3a, 2021 Q.3a, 2020 Q.2c **Rifat PDF: Pages 116–120 | Slides: Transport_Protocols_UDP_TCP_RTP.pdf**

Feature	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Guaranteed ordered delivery via ACKs	Best-effort, unreliable
Flow/Congestion	Sliding window + AIMD	None
Header Size	20 bytes minimum	8 bytes
Use Cases	FTP, HTTP, Email	DNS, VoIP, Gaming, Streaming

Q: TCP 3-Way Handshake & Congestion Control

Asked: 2022 Q.3a, 2021 Q.3a, 2020 Q.2c **Rifat PDF: Pages 119, 122 | Slides: Transmission_Control_Protocol_TCP.pdf**

3-Way Handshake: (1) Client sends **SYN** (seq=X). (2) Server replies **SYN-ACK** (seq=Y, ack=X+1). (3) Client sends **ACK** (ack=Y+1). Connection established.

Congestion Control (4 phases):

1. **Slow Start:** cwnd=1 MSS, doubles each RTT until threshold.
2. **Congestion Avoidance (AIMD):** After threshold, cwnd increases by +1 per RTT.
3. **Fast Retransmit:** 3 duplicate ACKs → retransmit lost segment immediately.
4. **Fast Recovery:** Halve threshold, resume from congestion avoidance (not slow start).

Q: Segment vs Packet vs Frame; Port vs Socket

Asked: 2020 Q.3a/b **Rifat PDF: Pages 32, 116**

- **Segment:** Layer 4 (Transport/TCP). Contains port numbers.
- **Packet/Datagram:** Layer 3 (Network/IP). Contains source/destination IP.
- **Frame:** Layer 2 (Data Link). Contains source/destination MAC + CRC trailer.
- **Port:** 16-bit number identifying a specific application (e.g., HTTP=80).
- **Socket:** IP Address + Port Number (e.g., `192.168.1.1:80`). Unique communication endpoint.

PART D: NETWORK LAYER

Q: IP Addressing — Classes, Classful vs Classless, Public vs Private

Asked: 2023 Q.4a/b, 2022 Q.3b/Q.4a, 2020 Q.4a, 2019 Q.4a–d **Rifat PDF: Pages 103–110 | Slides: IP_Addressing_Main_Slide.pdf, IPv4_Addresses_Classful_Addressing.pdf**

Class	Range	Default Mask	Purpose
A	0.0.0.0 – 127.255.255.255	/8	Very large networks
B	128.0.0.0 – 191.255.255.255	/16	Medium networks
C	192.0.0.0 – 223.255.255.255	/24	Small networks
D	224.0.0.0 – 239.255.255.255	—	Multicast

- **Classful:** Fixed class boundaries. **Classless (CIDR):** Variable-length prefix (e.g., /22).
- **Public IP:** Globally routable. **Private IP:** 10.x.x.x, 172.16–31.x.x, 192.168.x.x (not routable).
- **Subnet Mask:** 1s = network, 0s = host. **Wildcard Mask:** Inverse (used in ACLs).

Q: Subnetting & VLSM Calculations

Asked: 2023 Q.4c, 2020 Q.4b, 2019 Q.5b **Rifat PDF: Pages 106–109 | Slides: Subnetting_Supernetting_Classless_Addressing.pdf, Network_Season_07_VLSM.pdf**

Formula: Usable hosts = 2^h − 2. Find smallest h where 2^h − 2 ≥ requirement.

Solved: 172.16.2.1/22 for 50 PCs and 23 PCs (VLSM)

- **50 PCs:** Need 64 addresses (2^6). Prefix = /26, Mask = .192. Network: 172.16.0.0/26. Usable: .1–.62, Broadcast: .63
- **23 PCs:** Need 32 addresses (2^5). Prefix = /27, Mask = .224. Network: 172.16.0.64/27. Usable: .65–.94, Broadcast: .95

Solved: 210.210.210.0/24 for min 15 usable hosts

Need 2^5=32 (since 2^4−2=14 fails). Borrow 3 bits → /27. Creates 8 subnets of 30 usable hosts each.

Solved: 100.100.100.0/24 for min 30 usable hosts

Need 2^5=32 (30 usable). Borrow 3 bits → /27. Creates 8 subnets.

Q: IPv4 vs IPv6

Asked: 2023 Q.4b, 2022 Q.5b, 2020 Q.4c **Rifat PDF: Pages 59, 102, 112 | Slides: IP_Addressing_IPv4_and_IPv6.pdf**

Feature	IPv4	IPv6
Address Size	32-bit (4.3 billion)	128-bit (virtually infinite)
Header	Variable (20+ bytes with options)	Fixed 40-byte base header
Fragmentation	Routers can fragment	Only endpoints fragment (Path MTU Discovery)
NAT	Required (address exhaustion)	Not needed

Q: Subnetting vs Supernetting

Asked: 2023 Q.4b, 2022 Q.4b, 2020 Q.4a

- **Subnetting:** Borrow host bits → divide one network into smaller subnets.
- **Supernetting (Route Aggregation):** Borrow network bits → combine multiple networks into one supernetwork. Reduces routing table size.

Q: CIDR Routing Table Lookup

Asked: 2022 Q.5a

Given entries: 135.46.56.0/22→IF0, 135.46.60.0/22→IF1, 192.53.40.0/23→IF2, default→IF3.

- 135.46.63.10 → matches 135.46.60.0/22 → **Interface 1**
- 192.53.40.7 → matches 192.53.40.0/23 → **Interface 2**
- 135.46.57.14 → matches 135.46.56.0/22 → **Interface 0**
- 192.53.56.7 → no match → **Interface 3 (default)**

Q: ARP (Address Resolution Protocol)

Asked: 2022 Q.6a, 2021 Q.6c **Rifat PDF: Pages 1, 3 | Slides: Address_Resolution_Protocol_ARP.pdf**

Problem: Know the IP, need the MAC. **Solution:** Broadcast ARP Request ("Who has 192.168.1.5?"). Only the owner replies with its MAC via unicast ARP Reply.

Q: NAT (Network Address Translation)

Asked: 2023 Q.5b, 2022 Q.6b, 2019 Q.8b **Rifat PDF: Pages 89, 94 | Slides: Network_Address_Translation_NAT.pdf**

Router replaces private (Source IP, Port) with its public (NAT IP, New Port) in a translation table. Replies are looked up and forwarded back to the correct internal host.

Q: DHCP Process

Asked: 2023 Q.8b, 2019 Q.3d **Rifat PDF: Pages 126–127 | Slides: DHCP_Protocol_Breakdown.pdf**

DORA Process: (1) **Discover** (client broadcasts). (2) **Offer** (server proposes IP). (3) **Request** (client accepts). (4) **Acknowledge** (server confirms). Lease time limits how long the IP is held.

Q: Tunneling (IPv6 over IPv4)

Asked: 2022 Q.5b **Rifat PDF: Page 97 | Slides: IPv6_Extension_Headers_and_Tunneling.pdf**

IPv6 packet is encapsulated inside an IPv4 header to traverse legacy IPv4 networks. The IPv4 "Protocol" field is set to 41 (indicating IPv6 payload).

Q: Why IP is Connectionless; Transparent vs Non-Transparent Fragmentation

Asked: 2021 Q.4c, 2020 Q.6c, 2019 Q.8a **Rifat PDF: Page 99 | Slides: Transparent_vs_NonTransparent_Fragmentation.pdf**

IP is connectionless: No circuit setup before sending. Each packet routed independently. May arrive out of order. Reliability left to TCP.

- **Transparent:** Router fragments at entry to small-MTU network; exit router reassembles before forwarding. Outside world never knows.
- **Non-Transparent (IPv4 standard):** Router fragments; fragments stay split for the entire journey. Only final destination reassembles.

Q: MTU and Path MTU Discovery

Asked: 2023 Q.5c, Q.8c **Slides: DNS_Lookup_MTU_Discovery.pdf, Network_Season_10_Fragmentation.pdf**

MTU: Largest packet size a link can handle (Ethernet = 1500 bytes). **Path MTU Discovery:** Sender sets "Don't Fragment" (DF) bit. If a router can't forward it, it drops the packet and sends ICMP "Fragmentation Needed" with its MTU value. Sender shrinks packet size accordingly.

PART E: ROUTING & CONGESTION CONTROL

Q: Distance Vector vs Link State Routing

Asked: 2023 Q.6a, 2021 Q.5a, 2020 Q.5a, 2019 Q.6a **Rifat PDF: Pages 81–86 | Slides: Routing_Algorithms_Examples.pdf, Routing_Implementation_Part_1.pdf**

- **Distance Vector (Bellman-Ford, e.g., RIP):** Share entire table with neighbors only. Simple, low memory. Slow convergence, vulnerable to routing loops.
- **Link State (Dijkstra, e.g., OSPF):** Flood link status to ALL routers. Each builds a full topology map. Fast convergence, loop-free. Needs more CPU/RAM.

Q: Count-to-Infinity Problem

Asked: 2023 Q.6a, 2021 Q.5d, 2019 Q.6a **Rifat PDF: Page 81 | Slides: Routing_and_Congestion_Control.pdf**

When a link fails, DV routers keep incrementing hop counts based on stale neighbor info, bouncing packets in loops until the metric reaches "infinity" (16 in RIP). **Solutions:** Split Horizon, Route Poisoning.

Q: Default Route, Sink Tree, Route Poisoning, Split Horizon

Asked: 2023 Q.6b, 2022 Q.7b, 2021 Q.6a/b, 2020 Q.5b/c **Rifat PDF: Pages 78, 81 | Slides: Routing_Questions.pdf**

- **Default Route:** Forwarding rule used when no specific route matches. Packets go to default gateway.
- **Sink Tree:** Spanning tree of all shortest paths rooted at the destination.
- **Route Poisoning:** Advertising a dead route with metric=∞ to kill it immediately.

- **Split Horizon:** Never advertise a route back on the interface it was learned from.

Q: Dijkstra's Algorithm (Shortest Path)

Asked: 2023 Q.6b, 2021 Q.6a, 2020 Q.5b, 2019 Q.6b **Rifat PDF: Pages 84–86 | Slides: Routing_Algorithms_Examples.pdf**

Steps: (1) Set source distance=0, all others=∞. (2) Select unvisited node with lowest cost. (3) Update neighbors if new path is shorter. (4) Mark node visited. (5) Repeat until destination is reached. *(Specific numeric solutions depend on the graph figure provided in each exam paper — see Rifat PDF page 84 for the diagram format.)*

Q: Distance Vector Routing Table Calculation

Asked: 2023 Q.5b

Problem: Vectors into router C: from B:(5,0,8,12,6,2), from D:(16,12,6,0,9,10), from E:(7,6,3,9,0,4). Link costs: C-B=6, C-D=3, C-E=5. **Method:** For each destination, C's cost = min(cost_via_B, cost_via_D, cost_via_E).

- cost_via_X = link_cost(C,X) + X's distance to destination.

Q: Static vs Dynamic Routing; Intra-domain vs Inter-domain

Asked: 2021 Q.5d, 2020 Q.5a

- **Static:** Manually configured. Cannot adapt to failures. Good for tiny fixed networks.
- **Dynamic:** Routers auto-discover routes (RIP/OSPF/BGP). Adapts to link failures.
- **Intra-domain (IGP):** Routing within one AS (OSPF, RIP). Driven by performance.
- **Inter-domain (EGP):** Routing between AS's (BGP). Driven by policy.

Q: Hierarchical Routing

Asked: 2019 Q.6b

Addresses allocated in contiguous prefix chunks. Routers summarize massive blocks into single prefix routes using CIDR, reducing routing table size dramatically.

Q: Congestion Control — Leaky Bucket, Token Bucket, Choke Packets

Asked: 2023 Q.6c, 2021 Q.5b/c, 2020 Q.6a, 2019 Q.6c **Rifat PDF: Pages 88–93, 142 | Slides: QoS_Leaky_Token_Bucket_Loadshedding.pdf, Token_Bucket_vs_Leaky_Bucket.pdf**

- **Leaky Bucket:** Converts bursty traffic to fixed-rate output. Overflows = packets dropped. Strict policing.
- **Token Bucket:** Tokens added at fixed rate. Must spend token to send. Allows burst (up to bucket size). Drops excess *tokens*, not packets. **Better performance than Leaky Bucket.**
- **Hop-by-Hop Choke Packet:** Congested router sends choke signal to EVERY intermediate hop on the backward path, not just source. Provides immediate relief.

Q: Load Shedding and QoS

Asked: 2021 Q.6b **Slides: Load_Shedding_Wine_vs_Milk_Packets.pdf, Introduction_to_QoS.pdf**

- **Load Shedding:** Router drops packets when overwhelmed. May prioritize (e.g., keep newer "milk" packets over old "wine" packets).
- **QoS:** Strict priority queuing for time-sensitive traffic. Uses DiffServ/IntServ.

Q: Flow Control vs Congestion Control

Asked: 2023 Q.7a, 2022 Q.7a

- **Flow Control:** Point-to-point. Prevents fast sender from overwhelming slow receiver (advertised window).
- **Congestion Control:** Global. Prevents network saturation (congestion window). TCP effective window = min(cwnd, advertised window).

Q: Datagram vs Virtual Circuit

Asked: 2023 Q.5a, 2022 Q.7b **Rifat PDF: Pages 75, 90–91 | Slides: Datagram_vs_Virtual_Circuit_Packet_Switching.pdf**

- **Virtual Circuit:** Path established before data flows. Packets follow same route, arrive in order. Vulnerable to link failure.
- **Datagram:** Each packet independent, different paths possible. Robust to failure. May arrive out of order.
- **For low-latency trading:** Virtual Circuit recommended for consistent pathing.

PART F: APPLICATION LAYER & SECURITY

Q: DNS — Namespace and Resolver Lookup

Asked: 2021 Q.7a/b, 2020 Q.7a **Rifat PDF: Pages 123, 126–127 | Slides: Application_Layer_Domain_Name_System.pdf, DNS_Hierarchical_Architecture.pdf**

DNS Namespace: Inverted tree. Root → TLDs (.com, .bd) → Second-level (cuet.ac.bd). **Resolver Steps (MIT resolving cuet.ac.bd):**

1. Check local cache → 2. Query Root Server → returns .bd TLD server IP → 3. Query .bd → returns .ac.bd IP → 4. Query .ac.bd authoritative server → returns cuet.ac.bd IP → 5. Cache and return.

Q: Short Notes — SSH, FTP, SMTP, HTTP/HTTPS, WWW

Asked: 2022 Q.8c, 2021 Q.7c **Rifat PDF: Pages 4, 112 | Slides: Application_Layer_SMTP_and_FTP.pdf, HTTP_Fundamentals.pdf**

- **SSH (Port 22):** Encrypted remote terminal access, replaces Telnet.
- **FTP (Port 20/21):** Bulk file transfer between client/server.
- **SMTP (Port 25):** Push protocol for sending outgoing email.
- **HTTP (Port 80):** Web page request/response. **HTTPS (Port 443):** HTTP + TLS/SSL encryption.
- **WWW:** Global system of interlinked hypertext documents accessed via URLs.

Q: SMTP, POP3, IMAP — Email Protocols

Asked: 2023 Q.8a **Slides: Electronic_Mail_SMTP_POP_IMAP.pdf, How_Email_Works_SMTP_POP3_IMAP.pdf**

- **SMTP (Port 25):** Sending/relaying mail. Push only.
- **POP3 (Port 110):** Downloads mail, deletes from server. Bad for multi-device use.
- **IMAP (Port 143):** Syncs mail across all devices. Stays on server.

Q: ACL — Standard vs Extended

Asked: 2023 Q.7a, 2022 Q.8a, 2021 Q.6c/d, 2020 Q.5c/d, 2019 Q.5c **Rifat PDF: Pages 134–137 | Slides: Firewalls_and_Virtual_Private_Networks.pdf**

- **Standard ACL (1-99):** Filters by Source IP only. Place close to **destination**.
- **Extended ACL (100-199):** Filters by Source, Destination, Protocol, Port. Place close to **source** (saves bandwidth).

Q: Cryptography — Substitution & Transposition Ciphers

Asked: 2023 Q.7b, 2019 Q.7a **Rifat PDF: Page 138 | Slides: Network_Security_Module_8_Part_2.pdf**

- **Substitution (Caesar):** Replace each letter (e.g., shift 3: A→D). Vulnerable to frequency analysis.
- **Transposition:** Rearrange letter order using a keyword matrix. Letters unchanged, positions scrambled.

Q: Vigenère Cipher vs Caesar Cipher

Asked: 2022 Q.6c **Slides: Network_Security_Module_8_Part_2.pdf**

- **Caesar:** Mono-alphabetic, uniform shift. Easily broken by frequency analysis.
- **Vigenère:** Poly-alphabetic, variable shift per letter using a keyword. Flattens frequency distribution, making analysis much harder. **Yes, Vigenère is better.**

Q: DES Algorithm Principles

Asked: 2023 Q.7c, 2022 Q.8b, 2019 Q.7b **Rifat PDF: Page 129 | Slides: RSA_Cryptography.pdf**

DES: Symmetric block cipher. 64-bit blocks, 56-bit key, 16-round Feistel structure. Each round: expansion, XOR with round key, S-box substitution, P-box permutation. Now insecure due to short key; replaced by AES.

Q: RSA Algorithm — Encrypt "COMPUTER"

Asked: 2023 Q.7c, 2021 Q.8a/b **Rifat PDF: Page 129 | Slides: RSA_Cryptography.pdf**

RSA Steps: (1) Choose primes p, q. (2) n = p×q. (3) φ(n)=(p-1)(q-1). (4) Choose e coprime to φ(n). (5) d = e^{-1} mod φ(n). **Encrypt:** C = M^e mod n. **Decrypt:** M = C^d mod n. Convert each letter of "COMPUTER" to its numeric value, apply encryption formula.

Q: Transposition Cipher Problems (Solved)

Asked: 2023 Q.7c (QUESTION), 2021 Q.8c (DICTIONARY), 2020 Q.8b (LOCKDOWN), 2019 Q.7c (QUESTION), 2019 Q.8b (MEGABUCK)

Key="DICTIONARY", Plain="happinessistemporary"

Alphabetical order: A(8), C(3), D(1), I_1(2), I_2(5), N(7), O(6), R(9), T(4), Y(10) **Cipher:** `SAPEHSATIPERNOSRPMIY`

Key="LOCKDOWN", Plain="stayhomestaysafe"

Alphabetical: C(3), D(5), K(4), L(1), N(8), O_1(2), O_2(6), W(7) **Cipher:** `AAHSYYSSEETTOAMF`

Key="QUESTION", Plain="Weloveourcountrybangladesh"

Alphabetical: E(3), I(6), N(8), O(7), Q(1), S(4), T(5), U(2) **Cipher:** `LONXETAXUYEXORDXWRBSOUGXVNLXECAH`

Key="MEGABUCK", Plain="pleasetransferfivelatektoinmysonalibankaccount"

Alphabetical: A(4), B(5), C(7), E(2), G(3), K(8), M(1), U(6) **Cipher:** `AFAMBUSETYANTFKOKXLNEILCESLNIORITNAXPAVOACERESNT`

Q: Short Notes — ICMP, NAT, Digital Signature, FDDI, Firewall, VPN, VLAN

Asked: 2022 Q.8c, 2021 Q.7c, 2019 Q.8b/c **Rifat PDF: Pages 70–73 (VLAN/FDDI), 89 (NAT), 140 (VPN) | Slides: Various**

- **ICMP:** Error/diagnostic messages alongside IP (Ping, Traceroute, "Destination Unreachable").
- **Digital Signature:** Sender encrypts document hash with Private Key. Receiver verifies with sender's Public Key. Ensures non-repudiation.
- **Firewall:** Stateful security device between internal network and internet. Implements ACLs + deep packet inspection.
- **VPN:** Encrypted tunnel over public network for secure remote access.
- **VLAN:** Logical segmentation of a physical switch into multiple virtual subnets.

PART G: MODERN TOPICS (2023 New — Not in Rifat PDF)

Q: IoT, Cloud Computing, Autonomous Vehicles — Client-Server vs P2P

Asked: 2023 Q.4a, 2022 Q.1a

- **Cloud/IoT:** Client-Server model. Sensors/devices push data to central cloud servers.
- **Autonomous Vehicles:** Hybrid. Client-Server for map/traffic updates + P2P (Vehicle-to-Vehicle) for instant braking alerts.

Q: 5G Technology Features

Asked: 2023 Q.7a

(1) Up to 10 Gbps bandwidth. (2) ~1ms latency (URLLC). (3) 1M+ devices/km^2 (mMTC). (4) Network Slicing.

Q: Access Networks

Asked: 2023 Q.1b

The "last mile" connecting homes to ISP. Critical for bandwidth (symmetrical for video calls), latency (real-time), and scalability (massive simultaneous users).

Q: Cloud vs Quantum Computing

Asked: 2023 Q.3c

- **Cloud:** Classical bits (0 or 1). General-purpose, scalable via internet.

- **Quantum:** Qubits (superposition: 0, 1, or both). Exponentially faster for specific problems (e.g., breaking RSA).

Q: Cyber-Attacks Overview

Asked: 2023 Q.7b

(1) **DDoS:** Flood server with junk traffic. (2) **Phishing:** Trick humans into revealing passwords. (3) **MITM:** Intercept communication between two parties. (4) **Ransomware:** Encrypt victim's data, demand payment.

Q: Miscellaneous Comparisons

Asked across multiple years

Comparison	Key Difference
Hub vs Switch	Hub broadcasts; Switch forwards to specific port
Bridge vs Router	Bridge = L2 (MAC); Router = L3 (IP)
Internet vs Intranet	Public global vs. Private internal
Loopback vs Physical Address	127.0.0.1 (software test) vs MAC (hardware burned-in)
Message Switching vs Packet Switching	Whole message stored-and-forwarded vs. broken into packets pipelined
Connection-oriented vs Connectionless	TCP (handshake, reliable) vs UDP/IP (no setup, best-effort)
TP (Twisted Pair) vs Fiber	Copper, shorter distance vs. Glass, longer distance, higher speed
Data Link vs Network Layer	MAC/Framing (L2) vs. IP/Routing (L3)

PART H: ADDITIONAL QUESTIONS FOUND IN GAP ANALYSIS

> *These questions were identified as missing during the cross-reference audit.*

Q: What is Ethernet? What is Gigabit Ethernet?

Asked: 2021 Q.1b, 2019 Q.1c **Rifat PDF: Pages 61–62 | Slides: IEEE_802_3_Ethernet_Standard.pdf, Ethernet_CSMACD_Exponential_Backoff.pdf**

- **Ethernet (IEEE 802.3):** The dominant LAN technology. Defines wiring, signaling, and MAC addressing at Layers 1-2. Originally used CSMA/CD over coaxial bus; modern switched Ethernet eliminates collisions.
- **Gigabit Ethernet (IEEE 802.3z/ab):** Operates at 1000 Mbps (1 Gbps). Uses full-duplex point-to-point switched links. CSMA/CD is technically defined but practically not needed in full-duplex mode. Carrier Extension extends minimum frame size to 512 bytes for half-duplex Gigabit when needed.

Q: Which OSI Layer Handles What?

Asked: 2021 Q.2b **Rifat PDF: Pages 1–4**

Task	OSI Layer
Routing packets	Network Layer (Layer 3)
Running FTP and Telnet services	Application Layer (Layer 7)
Detecting and correcting errors	Data Link Layer (Layer 2)
Converting packets to frames	Data Link Layer (Layer 2)

Q: 5-Layer Protocol Hierarchy — Message Delivery Through Switch and Router

Asked: 2022 Q.2a **Rifat PDF: Pages 2–4 | Slides: Introduction_to_Layering.pdf**

A message of length `m` bytes at Application layer gets headers added at each layer:

```
Application:    [AH | m bytes]                  → m + h_a bytes
Transport:      [TH | AH | m]                   → m + h_a + h_t bytes
Network:        [NH | TH | AH | m]              → m + h_a + h_t + h_n bytes
Data Link:      [DH | NH | TH | AH | m | DT]    → + h_d + trailer
Physical:       Raw bits on wire

At a Link Layer Switch (Layer 2): Strips DH/DT → reads MAC → re-adds new DH/DT → forwards
At a Router (Layer 3): Strips DH/DT → reads NH (IP) → makes routing decision → adds new DH/DT
→ forwards
```

Q: Reasons for Using Layered Protocols

Asked: 2019 Q.2a **Rifat PDF: Pages 1–2**

1. **Modularity:** Each layer can be designed, tested, and updated independently.
2. **Abstraction:** Upper layers don't need to know the details of lower layers' implementation.
3. **Interoperability:** Standardized interfaces between layers allow equipment from different vendors to communicate.
4. **Ease of Maintenance:** A bug in one layer can be fixed without redesigning the entire protocol stack.
5. **Reusability:** Lower layers (Physical, Data Link) can be reused across many different applications.

Q: Hidden Station Problem

Asked: 2022 Q.3b **Slides: Wireless_MAC_Protocols.pdf**

Problem: In wireless networks, Station A and Station C can both communicate with Station B but are out of range of each other. A cannot "hear" C transmitting. Both may transmit to B simultaneously, causing a collision at B that neither A nor C can detect.

Solution: CSMA/CA with **RTS/CTS** handshake. Before transmitting, A sends a Request-To-Send (RTS) to B. B replies with Clear-To-Send (CTS) which is heard by ALL stations in B's range (including C). C now knows B is reserved and stays silent until A's transmission concludes.

Q: IPv6 Header Format

Asked: 2022 Q.6b **Rifat PDF: Page 102 | Slides: IPv6_Extension_Headers_and_Tunneling.pdf**

Fixed 40-byte IPv6 base header fields:

```
| Version (4b) | Traffic Class (8b) | Flow Label (20b)    |
| Payload Length (16b)  | Next Header (8b) | Hop Limit (8b)|
|             Source Address (128 bits)                    |
|           Destination Address (128 bits)                 |
```

- **Version:** Always 6. **Traffic Class:** QoS priority. **Flow Label:** Identifies a flow for special handling.
- **Next Header:** Replaces IPv4's "Protocol" field. Points to extension headers or upper layer (TCP=6, UDP=17).
- **Hop Limit:** Replaces TTL. Decremented by 1 at each router; packet discarded at 0.

Q: LAN vs WAN Differences

Asked: 2022 Q.2b **Slides: Introduction_to_WANs.pdf, WAN_Technology_Overview.pdf**

Feature	LAN	WAN
Coverage	Building/campus	State/country/global
Speed	1–100 Gbps	Typically ≤ 150 Mbps
Ownership	Organization-owned	Leased from ISPs/carriers
Cost	Low (internal cabling)	High (leased lines, satellites)
Error Rate	Very low	Higher due to distance

Q: Intranet vs Extranet

Asked: 2022 Q.1c **Slides: Enterprise_Networks_and_VPNs.pdf**

- **Intranet:** Private network strictly for internal employees. Behind corporate firewalls. Uses standard web technologies (HTTP, email) but only accessible inside.
- **Extranet:** Extension of the intranet granting controlled access to external trusted partners/vendors/customers via VPN or secure login.

Q: Simplex, Half-Duplex, Full-Duplex Communication

Asked: 2022 Q.1b **Slides: Data_Communication_Fundamentals.pdf**

Mode	Direction	Example
Simplex	One-way only	TV broadcast, Keyboard→CPU

Half-Duplex	Two-way, one at a time	Walkie-talkie
Full-Duplex	Two-way, simultaneous	Telephone, modern Ethernet

Q: Centralized vs Distributed Networks

Asked: 2022 Q.2c

- **Centralized:** All users connect to one central server. Single point of failure. Simple but non-scalable.
- **Distributed:** Multiple cooperating servers/nodes. No single point of failure. Highly scalable and robust.

Q: Message Switching vs Packet Switching

Asked: 2022 Q.8a **Slides: Packet_Switching_in_WAN.pdf, Switched_Communication_Networks_Part_1.pdf**

Feature	Message Switching	Packet Switching
Data Unit	Entire message as one block	Message broken into small packets
Storage	Intermediate nodes store full message (huge memory)	Nodes buffer small packets (fast memory)
Delay	Very high (must receive entire message before forwarding)	Low (pipelining — forward packets as they arrive)
Channel Usage	Blocks the link for the full message duration	Multiple users interleave packets on same link

---

Q: Frame Relay and ATM

Asked: 2022 Q.8b **Slides: atm datagram virtual ckt.pdf, X25_Virtual_Circuits_DTE_DCE_PSE.pdf**

Frame Relay: Packet-switched WAN technology. **Unreliable by design** — no ACKs, no retransmission at Layer 2. Assumes clean fiber lines. Error recovery pushed to TCP. Simple and fast but drops erroneous frames silently.

ATM (Asynchronous Transfer Mode): Uses small fixed-size "cells" (53 bytes: 5-byte header + 48-byte payload). High overhead ratio. Excellent QoS but expensive hardware. Largely replaced by MPLS.

Q: EGP vs OSPF

Asked: 2022 Q.8c

Feature	EGP/BGP (Exterior)	OSPF (Interior)
Scope	Between Autonomous Systems	Within one AS
Algorithm	Path Vector (BGP)	Link State (Dijkstra)
Driven By	Policy, cost, agreements	Performance metrics (link cost)
CIDR Support	BGP fully supports	Fully supports

Q: Connection-Oriented vs Connectionless Service

Asked: 2023 Q.5a, 2022 Q.5a **Slides: Datagram_vs_Virtual_Circuit_Packet_Switching.pdf**

Feature	Connection-Oriented (TCP)	Connectionless (UDP/IP)
Analogy	Telephone call	Postal system
Setup	3-way handshake required	No setup, send directly
Ordering	Guaranteed in-order delivery	May arrive out of order
Reliability	ACKs, retransmission	Best-effort, no guarantees
Robustness	Vulnerable to path failure	Routes around failures

Q: CRC with Generator $G(x) = x^5 + x + 1$

Asked: 2022 Q.2c

- Generator Binary: x^5+x+1 → `100011` (degree 5)
- Input frame: `101101110011`
- Append 5 zeros: `10110111001100000`
- Perform modulo-2 division by `100011`
- The 5-bit remainder is the CRC checksum
- Transmitted frame = original + remainder

Q: Noiseless Channel Bit Rate Calculation (Nyquist)

Asked: 2022 Q.2b

Problem: Signal through noiseless channel, capacity 128 Kbps max. Using Nyquist: $C = 2 \times B \times \log_2(L)$. If binary (L=2): $128000 = 2 \times B \times 1$ → **B = 64,000 Hz = 64 KHz**

Q: HTTP vs HTTPS, Telnet vs SSL, FTP vs HTTP, Port Numbers

Asked: 2019 Q.3a–d **Slides: HTTP_Fundamentals.pdf, Hyper_Text_Transfer_Protocol_HTTP.pdf**

Protocol	Port	Description
HTTP	80	Unencrypted web page transfer
HTTPS	443	HTTP + TLS/SSL encryption
FTP	20/21	File transfer (data/control channels)
Telnet	23	Unencrypted remote terminal (insecure)
SSH	22	Encrypted remote terminal (replaces Telnet)
SMTP	25	Sending email
DNS	53	Domain name resolution

POP3	110	Downloading email
IMAP	143	Syncing email

- **HTTP vs HTTPS:** HTTPS adds TLS/SSL encryption layer. Validates server identity via certificates.
- **Telnet vs SSH:** Both provide remote terminal. Telnet sends plaintext passwords; SSH encrypts everything.
- **FTP vs HTTP:** FTP optimized for bulk file transfer (resume, binary mode). HTTP optimized for stateless web page requests.

Q: IP Datagram Format (IPv4 Header)

Asked: 2021 Q.6d **Rifat PDF: Page 99 | Slides: Internet_Protocol_Router_Forwarding.pdf**

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|Version|  IHL  |Type of Service|          Total Length         |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|         Identification        |Flags|     Fragment Offset     |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  Time to Live |    Protocol   |       Header Checksum         |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                       Source IP Address                       |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                    Destination IP Address                     |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

Minimum header = 20 bytes. **TTL** = max hops before discard. **Protocol** = upper layer (TCP=6, UDP=17).

Q: ACL vs Firewall

Asked: 2020 Q.5c/d

- **ACL:** Simple ordered rule list on router interfaces. Stateless — examines each packet in isolation. Filters Layer 3-4 (IP/Port).
- **Firewall:** Full security appliance. **Stateful** — tracks TCP connection state. Deep Packet Inspection (Layer 7). Sits between internal network and internet. Implements ACLs as one of many features.

Q: Flow Control and Error Control Methods

Asked: 2019 Q.4d **Rifat PDF: Pages 54–55 | Slides: Data_Link_Flow_and_Error_Control.pdf**

Flow Control: Regulates sender speed to prevent overwhelming the receiver.

- **Stop-and-Wait:** Send 1 frame, wait for ACK. Simple but wasteful.
- **Sliding Window:** Send W frames before waiting. Efficient.

Error Control: Detects and corrects errors.

- **Parity Check:** Single bit added. Detects odd number of errors.
- **CRC:** Polynomial division. Detects burst errors.

- **Hamming Code:** Locates and corrects single-bit errors.

Q: 127.0.0.0 (Loopback Address)

Asked: 2019 Q.4a

127.0.0.0/8 is the **loopback** address block. 127.0.0.1 is used to test the local TCP/IP stack without sending packets onto the actual network. Any packet sent to this address is looped back internally by the OS.

PART I: FINAL AUDIT — REMAINING GAPS

Q: TCP Connection Termination (4-Way Handshake)

Asked: 2022 Q.3a (terminate) **Rifat PDF: Pages 119–120 | Slides: Transmission_Control_Protocol_TCP.pdf**

TCP connection is **terminated** using a 4-way handshake (FIN exchange):

1. **FIN (Client→Server):** Client sends FIN segment, enters FIN-WAIT-1.
2. **ACK (Server→Client):** Server acknowledges FIN, enters CLOSE-WAIT. Client enters FIN-WAIT-2.
3. **FIN (Server→Client):** Server finishes sending data, sends its own FIN.
4. **ACK (Client→Server):** Client acknowledges Server's FIN, enters TIME-WAIT (waits 2×MSL), then CLOSED.

Q: Routing vs Forwarding

Asked: 2021 Q.5a **Rifat PDF: Pages 78, 81**

- **Routing (Control Plane):** The global, distributed process of calculating optimal end-to-end paths across the entire network. Builds routing tables. Runs algorithms like Dijkstra or Bellman-Ford.
- **Forwarding (Data Plane):** The fast, local action at a single router. When a packet arrives, the router looks up the destination IP in its forwarding table and switches the packet to the correct output port. Happens in hardware at wire speed.

Q: Link State Routing Algorithm — Steps

Asked: 2022 Q.7a **Rifat PDF: Pages 84–86 | Slides: Routing_Implementation_Part_2.pdf**

Steps of Link State Routing:

1. **Discover Neighbors:** Each router discovers its directly connected neighbors and their link costs.
2. **Build LSP (Link State Packet):** Each router constructs a packet containing its identity, sequence number, age, and list of neighbors with costs.
3. **Flood LSP:** Each router distributes its LSP to ALL other routers in the network (not just neighbors).
4. **Build Topological Map:** Each router independently assembles a complete graph of the entire network from all received LSPs.
5. **Run Dijkstra:** Each router independently computes the shortest path tree from itself to every other router using Dijkstra's algorithm.

Q: Prominent Factors/Causes of Congestion

Asked: 2023 Q.6c, 2020 Q.6a, 2019 Q.6c **Rifat PDF: Pages 88–89 | Slides: Congestion_Control_in_Routing.pdf**

Congestion occurs when total packets injected exceed network carrying capacity:

1. **Insufficient bandwidth:** Too many hosts pumping data beyond link capacity.
2. **Slow router processors:** Slow switching fabrics bottleneck queued packets.
3. **Inadequate buffer memory:** Router queues overflow, causing packet drops.
4. **Mismatched line speeds:** A fast LAN dumping into a slow WAN.
5. **Bursty traffic patterns:** Multiple hosts burst simultaneously.

Q: UTP vs STP (Twisted Pair)

Asked: 2023 Q.1b **Rifat PDF: Pages 58–59 | Slides: Physical_Layer_Transmission_Media.pdf**

- **UTP (Unshielded Twisted Pair):** Cheaper, thinner, easier to install. Adequate for most indoor LANs (Cat5e/Cat6). No grounding required.
- **STP (Shielded Twisted Pair):** Has foil/braided shielding. Better EMI protection for industrial environments. More expensive, bulkier, must be properly grounded (improper grounding makes shield act as antenna).

Q: Star-Ring Hybrid Topology

Asked: 2023 Q.1c **Slides: LAN_Technology_and_Network_Topologies.pdf**

Hybridizing Star + Ring: Physically wired as a Star (nodes connect to central MAU — Multistation Access Unit), but logically data flows in a Ring. The MAU internally connects each node's transmit to the next node's receive.

Setup Challenges: (1) Central MAU = single point of failure. (2) Need hardware to auto-bypass faulty nodes. (3) Requires separate TX/RX pairs from each node to hub.

Q: Extended Star Topology vs Standard Star

Asked: 2023 Q.2a

- **Standard Star:** All nodes connect to ONE central hub/switch.
- **Extended Star:** Multiple star sub-networks connected hierarchically via a backbone. Increases coverage area and compartmentalizes traffic.
- **Disadvantages of Star:** (1) Central point of failure. (2) Cable cost proportional to number of nodes.

Q: Client-Server vs Peer-to-Peer Model; Computer vs Autonomous System

Asked: 2023 Q.4a, 2022 Q.1a, 2021 Q.1a **Rifat PDF: Pages 1–3**

Client-Server: Centralized. Thin clients request from powerful servers. Easy to manage. Examples: Web browsing, Email, Cloud apps. **Peer-to-Peer (P2P):** Decentralized. Every node acts as both client and server. No central dependency. Examples: BitTorrent, Bitcoin.

Computer vs Autonomous System (AS):

- **Computer:** A single end-device (host) on a network.
- **Autonomous System (AS):** A set of routers under single administrative control, using one IGP internally, presenting a unified routing policy externally via BGP.

Q: IPv4 Packet Fragmentation Calculation (MTU = 500 bytes)

Asked: 2023 Q.8c **Rifat PDF: Page 99 | Slides: Network_Season_10_Fragmentation.pdf**

Method for fragmenting an IPv4 packet when outgoing MTU = 500 bytes:

1. **Max data per fragment** = MTU - IP Header = 500 - 20 = 480 bytes.
2. **Fragment offset unit** = 8 bytes. So max data must be a multiple of 8: 480 ÷ 8 = 60 (exact, good).
3. **Number of fragments** = ⌈Total Data / 480⌉
4. Each fragment gets: same Identification field, MF (More Fragments) flag set on all but the last, Fragment Offset = (previous fragments' total data) / 8.

Example: Original packet = 4000 bytes total (3980 data + 20 header).

- Fragment 1: Offset=0, Data=480 bytes, MF=1
- Fragment 2: Offset=60, Data=480 bytes, MF=1
- ... continue until all data is covered
- Last Fragment: MF=0

Q: Why Do We Need Cryptography?

Asked: 2023 Q.7b **Rifat PDF: Pages 129, 134**

Reasons:

1. **Confidentiality:** Prevent unauthorized reading of sensitive data.
2. **Integrity:** Detect if data has been tampered with during transmission.
3. **Authentication:** Verify the identity of the sender.
4. **Non-repudiation:** Sender cannot deny having sent a message (via digital signatures).

Q: "Explicitly Deny" in ACL

Asked: 2023 Q.7a

Every ACL has an **implicit deny all** at the end — any packet that doesn't match any rule is silently dropped. An **explicit deny** is when the administrator deliberately writes a deny rule (e.g., `access-list 10 deny host 192.168.1.100`) to specifically block certain traffic and optionally log it, making the denial visible and auditable rather than silent.

Q: DHCP Leasing and Renewal

Asked: 2023 Q.8b **Rifat PDF: Pages 126–127 | Slides: DHCP_Protocol_Breakdown.pdf**

- **DHCP Lease:** An IP address is assigned for a limited time period (lease time), not permanently. This ensures addresses return to the pool when devices disconnect.
- **DHCP Renewal:** At 50% of lease time (T1), client sends DHCPREQUEST to renew. If no response, at 87.5% (T2) it broadcasts to any DHCP server. If lease expires without renewal, client must restart the DORA process.

> ***END OF WRITTEN ANSWERS — Diagram Appendix follows below.***

📐 APPENDIX: MUST-KNOW DIAGRAMS (Hand-Draw These in Exam)

> *Every diagram below has been asked in exams. Practice sketching them.* ***Page numbers refer to the Rifat PDF.***
> *Slide files are in* `materials/slides/` *.*

Diagram 1: Network Topologies

When asked: 2023 Q.1c, 2022 Q.2a, 2021 Q.1b, 2020 Q.1b | **Rifat PDF: Pages 11, 14–15 Slide:** `Network_Topology_Overview.pdf` , `LAN_Technology_and_Network_Topologies.pdf`

```
BUS:                STAR:                 RING:
 ═══┬═══┬═══┬═══    [PC]  [PC]            [PC]──[PC]
    │   │   │          \   |   /              |       |
  [PC] [PC] [PC]       [HUB]               [PC]──[PC]
                       /   |   \
                    [PC]  [PC]  [PC]

MESH (Full):          TREE/EXTENDED STAR:
[PC]──[PC]                [Core Switch]
 |  ╲╱  |                /     |      \
[PC]──[PC]          [Switch] [Switch] [Switch]
                    / | \    / | \     / | \
                  PCs PCs  PCs PCs   PCs PCs
```

Diagram 2: OSI vs TCP/IP Layers

When asked: 2023 Q.1a, 2020 Q.2a, 2019 Q.1a | **Rifat PDF: Pages 1, 4 Slide:** `OSI_and_TCP_IP_Layering_Models.pdf`

```
OSI Model (7)          TCP/IP Model (4)
┌──────────────┐       ┌──────────────┐
│ Application  │       │              │
├──────────────┤       │ Application  │
│ Presentation │       │              │
├──────────────┤       │              │
│   Session    │       │              │
├──────────────┤       ├──────────────┤
│  Transport   │       │  Transport   │
├──────────────┤       ├──────────────┤
│   Network    │       │   Internet   │
├──────────────┤       ├──────────────┤
│  Data Link   │       │              │
├──────────────┤       │Network Access│
```

```
│   Physical   │      │              │
└──────────────┘      └──────────────┘
```

Diagram 3: Encapsulation Process

When asked: 2023 Q.1a, 2022 Q.2a | **Rifat PDF: Pages 2–4 Slide:** `Introduction_to_Layering.pdf`

```
Application:  [     DATA      ]
Transport:    [TCP Hdr][  DATA  ]          ← Segment
Network:      [IP Hdr][TCP Hdr][DATA]      ← Packet
Data Link:    [MAC][IP Hdr][TCP][DATA][CRC]  ← Frame
Physical:     101010110011100101...          ← Bits
```

Diagram 4: TCP 3-Way Handshake

When asked: 2022 Q.3a, 2021 Q.3a, 2020 Q.2c | **Rifat PDF: Pages 119–120 Slide:** `Transmission_Control_Protocol_TCP.pdf`

```
Client                        Server
  |                              |
  |──── SYN (seq=X) ─────────────>|
  |                              |
  |<──── SYN-ACK (seq=Y, ack=X+1)─|
  |                              |
  |──── ACK (ack=Y+1) ────────────>|
  |                              |
  |====== CONNECTION ESTABLISHED ==|
```

Diagram 5: DHCP DORA Process

When asked: 2023 Q.8b | **Rifat PDF: Pages 126–127 Slide:** `DHCP_Protocol_Breakdown.pdf`, `DHCP_and_DNS_Lecture_52.pdf`

```
Client                          DHCP Server
  |                                 |
  |── (1) DHCP DISCOVER (broadcast) ->|
  |                                 |
  |<── (2) DHCP OFFER (IP proposed) ──|
  |                                 |
  |── (3) DHCP REQUEST (accept) ──────>|
  |                                 |
  |<── (4) DHCP ACK (confirmed) ──────|
  |                                 |
  |   [Client now uses assigned IP]   |
```

Diagram 6: ARP Process

When asked: 2022 Q.6a | **Rifat PDF: Page 3 Slide:** `Address_Resolution_Protocol_ARP.pdf`, `arp lec51.pdf`

```
Host A (knows IP, needs MAC)           Host B (target)
   |                                       |
   |── ARP Request (broadcast) ──────────> | (all hosts)
   |   "Who has 192.168.1.5?"             |
   |                                       |
   |<── ARP Reply (unicast) ──────────────|
   |   "I am 192.168.1.5, MAC=AA:BB:..."  |
   |                                       |
   [A caches B's MAC in ARP table]
```

Diagram 7: DNS Resolver Lookup

When asked: 2021 Q.7a, 2020 Q.7a | **Rifat PDF: Pages 123, 126 Slide:** `DNS_Hierarchical_Architecture.pdf`, `DNS_Lookup_Caching_DHCP_ICMP.pdf`

```
[Client PC] ──> [Local DNS Server]
                      |
              (1)     |──> [Root DNS Server (.)]
              (2)     |<── "Go ask .bd TLD server"
              (3)     |──> [.bd TLD Server]
              (4)     |<── "Go ask .ac.bd server"
              (5)     |──> [.ac.bd Authoritative]
              (6)     |<── "cuet.ac.bd = 203.0.113.5"
                      |
[Client PC] <── [Returns IP + Caches it]
```

Diagram 8: NAT Translation

When asked: 2023 Q.5b, 2022 Q.6b | **Rifat PDF: Pages 89, 94 Slide:** `Network_Address_Translation_NAT.pdf`

```
Private Network              NAT Router               Internet
┌──────────────┐         ┌──────────────────┐      ┌──────────┐
│192.168.1.50 │──────> │ Replaces src IP  │─────>│          │
│   :3211      │         │ 192.168.1.50:3211│      │ Web      │
└──────────────┘         │ with             │      │ Server   │
                         │ 203.0.113.5:60001│      │104.21.3.4│
┌──────────────┐         │                  │      │          │
│192.168.1.65 │──────> │ Translation Table│      └──────────┘
│   :5432      │         │ maps both ways   │
└──────────────┘         └──────────────────┘
```

Diagram 9: IPv6 Tunneling Through IPv4

When asked: 2022 Q.5b | **Rifat PDF: Page 97 Slide:** `IPv6_Extension_Headers_and_Tunneling.pdf`

```
[IPv6 Host] ──> [Dual-Stack Router A] ═══════════> [Dual-Stack Router B] ──> [IPv6 Host]
                        |        IPv4 Network              |
                        |                                  |
                  Encapsulate:                       Decapsulate:
```

```
                    [IPv4 Hdr][IPv6 Pkt]          Remove IPv4 Hdr
                    Protocol Field = 41           Deliver IPv6 Pkt
```

Diagram 10: Leaky Bucket vs Token Bucket

When asked: 2023 Q.6c, 2021 Q.5b | **Rifat PDF: Pages 88, 92, 142 Slide:** `Token_Bucket_vs_Leaky_Bucket.pdf`, `QoS_Leaky_Token_Bucket_Loadshedding.pdf`

```
LEAKY BUCKET:                     TOKEN BUCKET:
 Bursty Input                      Bursty Input
     ↓                                 ↓
 ┌────────┐                        ┌────────┐
 │ Bucket │ ← overflow=DROP        │ Queue  │ ← packets wait
 │        │                        └───┬────┘
 └───┬────┘                            │ need token to send
     │ fixed rate output             ┌───┴────┐
     ↓                               │ Token  │ ← tokens added
 ═══════                             │ Bucket │   at fixed rate
 Constant                            └───┬────┘
 Rate                                    │
                                         ↓
                                   Bursty OK (up to bucket size)
```

Diagram 11: ACL Flowchart

When asked: 2022 Q.8a | **Rifat PDF: Pages 134–137 Slide:** `Firewalls_and_Virtual_Private_Networks.pdf`

```
Incoming Packet
     │
     ▼
┌──────────────┐    YES     ┌──────────┐
│ Match Rule 1?│──────────> │ Permit/  │
└──────┬───────┘            │ Deny     │
       │ NO                 └──────────┘
       ▼
┌──────────────┐    YES     ┌──────────┐
│ Match Rule 2?│──────────> │ Permit/  │
└──────┬───────┘            │ Deny     │
       │ NO                 └──────────┘
       ▼
┌─────────────────┐
│ Implicit Deny All│  ← DEFAULT: drop packet
└─────────────────┘

Standard ACL: Place near DESTINATION
Extended ACL: Place near SOURCE
```

Diagram 12: Transparent vs Non-Transparent Fragmentation

When asked: 2021 Q.4c, 2019 Q.8a | **Rifat PDF: Page 99 Slide:** `Transparent_vs_NonTransparent_Fragmentation.pdf` , `Network_Season_10_Fragmentation.pdf`

```
TRANSPARENT:
[Src]──>[G1: Fragment]──>[Small MTU Net]──>[G2: Reassemble]──>[Dest]
                                          ^ reassembled HERE

NON-TRANSPARENT (IPv4):
[Src]──>[Router: Fragment]──>[Net A]──>[Net B]──>[Dest: Reassemble]
                                                  ^ reassembled HERE only
```

Diagram 13: Transposition Cipher Matrix (Column Method)

When asked: 2023, 2021, 2020, 2019 | **Rifat PDF: Page 138**

```
Key:  D  I  C  T  I  O  N  A  R  Y
      1  2  3  4  5  6  7  8  9  10
Row1: h  a  p  p  i  n  e  s  s  i
Row2: s  t  e  m  p  o  r  a  r  y

Step: Sort key alphabetically → read columns in that order.
A=col8, C=col3, D=col1, I=col2, I=col5, N=col7, O=col6, R=col9, T=col4, Y=col10
```

Diagram 14: FDDI Dual Ring

When asked: 2021 Q.1d, 2019 Q.8b | **Rifat PDF: Page 73 Slide:** `LAN_Technologies_Must_Read.pdf`

```
        ┌──[Station A]──┐
        │   ↓ Primary    │
   [Station D]      [Station B]
        │   ↑ Secondary  │
        └──[Station C]──┘

If cable breaks between A & B:
Rings "wrap" → Primary + Secondary form one big loop
```

> ***END OF MASTER DOCUMENT — All unique questions from 2019–2023 solved. Total: 60+ topics | 12+ numerical solutions | 15+ comparison tables | 14 hand-drawable diagrams***